MIS207 · RESEARCH METHODS

Introduction to Research

What counts as knowledge, how we justify a claim, and why the method you choose always follows from what you believe reality is.



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Today

01

Ways of knowing

Why science is only one route to knowledge and what makes it different.

02

What research is

Kerlinger's definition and the three properties every study must have.

03

Ontology · Epistemology · Axiology

The three questions you answer before you touch a method.

04

Paradigms

Positivist, interpretive, critical, pragmatic and Kuhn's paradigm shifts.

05

Research in 2026

Open science, the replication crisis, and AI in the research pipeline.

06

Lab

Google Scholar and the İDU Library system — hands-on.

Why methods matter more, not less, in 2026

Tools that write fluent text are now free and everywhere. That makes the ability to judge whether a claim is any good the scarce skill and the one this course is about.

Five ways of knowing

Science is not the only way to know something. It is the one that builds in a way of catching its own mistakes.

Route	What it gives you	Where it breaks down
Experience	Direct, vivid, personally convincing evidence.	Unsystematic and unrepresentative; we remember the striking case, not the typical one.
Authority	Fast access to accumulated expertise.	Experts disagree, and are wrong in ways you cannot audit from the outside.
Tradition / consensus	Shared knowledge that coordinates a community.	Popularity is not evidence; it preserves errors as faithfully as truths.
Logical reasoning	Valid conclusions from stated premises.	A perfectly valid argument from false premises is still false.
Science	Claims that are testable, public and correctable.	Slow, expensive, provisional — and only as good as the method behind it.

A DEFINITION OF RESEARCH

The systematic, controlled, empirical and critical investigation of hypothetical propositions about the presumed relations among natural phenomena.

Kerlinger (1970), on what research is — as discussed in Cohen, Manion & Morrison (2018)

Three properties every study must have

Research is a combination of experience and reasoning and, crucially, of a community that checks it.

01

Systematic and Controlled

Experience delivers events haphazardly. Research works on the inductive–deductive model: concepts and procedures are defined precisely enough that someone else could repeat what you did.

02

Empirical

The researcher turns to experience for validation. A claim that no possible observation could challenge is not yet a research claim.

03

Self-correcting

Procedures and results are open to public scrutiny by other researchers. Incorrect results are, in time, found and either revised or discarded (Mouly, 1978).

Three questions you answer before choosing a method

Method is the last decision, not the first. These three answers constrain everything that follows.

WHAT EXISTS

Ontology

What is the nature of reality? Is the social world out there, independent of us and waiting to be observed — or is it produced by the people in it? Objectivism vs. constructionism.

HOW WE KNOW

Epistemology

What counts as acceptable knowledge, and how do we get it? Hard, observable, measurable facts — or meanings that have to be interpreted from the inside? Positivism vs. interpretivism.

WHAT MATTERS

Axiology

What is the role of values? Does the researcher stand outside the study — or are their values part of it, to be declared rather than denied?

KUHN (1962)

Paradigms: the shared assumptions of a research community

— **A paradigm is** a way of looking at or researching phenomena — a world view, an 'accepted model or pattern' (Kuhn, 1962), a shared belief system and the identity of a research community.

— **It settles in advance** which problems count as worth investigating, which methods are legitimate, and what an acceptable solution looks like.

— **It is not just methodology** (Hammersley, 2013). Paradigms are assumptions about what the world is like and how we can know it — method follows from them.

— **Paradigms shift** When an existing paradigm can no longer explain a phenomenon as well as a rival, a scientific revolution occurs and the new paradigm becomes the 'normal science' of the day.

— **Why you care** Reviewers judge your work from inside a paradigm. Stating yours explicitly is how you stop a methodological disagreement from being read as an error.

The Map of Paradigms

Paradigm	What is reality?	How do we know?	Typical methods
Positivist	One reality, external and measurable.	Objective observation; test hypotheses; seek general laws.	Experiments, surveys, tests
Post-positivist	One reality, but only imperfectly knowable.	Falsification, not proof; all findings are provisional and fallible.	Quasi-experiments, modelling
Interpretive	Multiple realities, constructed by people.	Understand meaning from the participant's standpoint (verstehen).	Interviews, ethnography, case study
Critical	Reality shaped by power, history and ideology.	Expose the structures behind what looks natural; aim to change it.	Action research, critical discourse analysis
Pragmatic	Whatever works for the problem at hand.	The research question decides; combine what each approach does best.	Mixed methods

Positivism

What it assumes

· **Determinism** events have causes, and those causes can be uncovered.

· **Empiricism** reliable knowledge derives only from experience.

· **Parsimony** the simplest explanation that fits the data wins — Occam's razor.

· **Generality** from observations of the particular, generalise to the whole.

· **Objectivity** the researcher is an observer, standing outside the phenomenon.

Where it struggles

· **Human complexity** social phenomena are elusive and intangible next to the regularity of the natural world.

· **Meaning** behaviour that is identical on a questionnaire can mean opposite things to two people.

· **Context stripped** controlling variables removes exactly the setting that explains the behaviour.

· **Value-free is a claim** choosing what to measure is already a value judgement.

· **Dehumanising** treating people as passive respondents ignores their agency.

The Positivist Research Cycle

Note the loop: 'seek alternative explanations' sends you back to the start. The cycle never actually ends.

01
Identify topic
A puzzle worth the effort.

02
Review literature
What is already known, and what is missing.

03
Formulate hypothesis
A testable statement about a relation.

04
Design research
Sample, instruments, procedure, controls.

05
Collect data
Systematically, as designed.

06
Analyse data
Does the evidence support the hypothesis?

07
Seek alternatives
What else could explain this result?

08
Report findings
Publicly, so others can check and extend.

The Interpretive Alternative

If reality is produced by people, the researcher's job is to understand it from the inside rather than measure it from the outside.

01

Phenomenology

Studies conscious experience: how does this look and feel from inside the person living it? Brackets the researcher's assumptions.

02

Ethnomethodology

Studies the ordinary methods people use to produce social order — the unwritten rules that make a meeting or a queue work.

03

Symbolic interactionism

Society is made in interaction. People act toward things on the basis of the meanings those things have for them.

04

Constructionism

Knowledge and categories are built socially and historically, not discovered ready-made in the world.

Three Approaches

	Quantitative	Qualitative	Mixed methods
Asks	How much? How many? How strongly related?	How and why? What does it mean to them?	Both, deliberately combined.
Logic	Deductive: theory → hypothesis → test.	Inductive: data → patterns → theory.	Abductive: iterate between the two.
Data	Numbers: scales, counts, logs, instruments.	Words, images, observations, documents.	Both strands, integrated at a defined point.
Sampling	Large, probability-based, aiming to generalise.	Small, purposive, aiming for depth.	Sequential or parallel, by design.
Quality judged by	Validity, reliability, generalisability.	Credibility, transferability, dependability, confirmability.	Integration quality and inference transferability.

01

Research Today

What has changed in the last decade and what it means for the study you are about to design.

The Open Science Turn

01

Preregistration

You publish your hypotheses and analysis plan before collecting data, so the line between prediction and post-hoc story stays visible. Registered Reports go further: peer review happens before results exist.

02

Open data & FAIR

Data deposited so it is Findable, Accessible, Interoperable and Reusable. Increasingly a condition of funding, not a courtesy.

03

Preprints & open access

arXiv, SSRN and institutional repositories put work in public before or alongside journal publication — faster, but unrefereed until it isn't.

04

Reproducible workflows

Scripted analysis (R, Python) and version control, so that 'how did you get this number?' has an answer that is not a spreadsheet nobody can find.

A DISCIPLINE CORRECTING ITSELF

The Replication Crisis

What happened

- **Ioannidis (2005)** argued on statistical grounds that most published research findings are false.
- **Open Science Collaboration (2015)** replicated 100 psychology studies: 97% of originals were significant, 36% of replications were.
- **The causes are structural** small samples, flexible analysis, publishing only what 'worked', and no reward for checking.
- **It is not confined to psychology** similar findings have followed in medicine, economics and management.

What it changes for your project

- **Say what you did** a methodology section that cannot be followed cannot be checked.
- **Report the whole picture** including what did not work; a null result is a result.
- **Beware p-hacking** testing until something is significant is not analysis, it is searching.
- **Prefer estimates to verdicts** effect sizes and intervals say more than a bare 'significant / not significant'.
- **Scepticism is professional** doubting a finding is not rudeness; it is the job.

USE IT AND AUDIT IT

AI in the Research Pipeline

Where it genuinely helps

- **Discovery** semantic search surfaces papers keyword search misses — Elicit, Consensus, Semantic Scholar, Connected Papers.
- **Orientation** fast summaries of an unfamiliar field, to be verified, not cited.
- **Language** editing and clarity support for writing in a second language.
- **Code and analysis** drafting scripts, explaining an error, suggesting a test.
- **Structured extraction** pulling sample sizes or measures out of many papers into one table.

Where it fails, predictably

- **Fabricated citations** 55% of GPT-3.5 and 18% of GPT-4 references in short reviews were invented (Walters & Wilder, 2023).
- **Confident wrongness** fluency is not accuracy, and the output carries no signal of its own reliability.
- **Stale or thin coverage** training data has a cutoff; paywalled and non-English work is under-represented.
- **Hidden bias** it reproduces whatever is dominant in its training data.
- **Judgement** it cannot decide whether your research question is worth asking.

COURSE POLICY

Using AI Honestly in This Course

The standard is simple: you must be able to defend every sentence you submit as your own reasoning.

— **Allowed** Searching, summarising for your own understanding, language editing, help with code, and explaining concepts you are stuck on.

— **Not allowed** Submitting generated text as your own analysis, or citing a source you have not opened and read.

— **Always verify** Every citation must be traced to the actual article — title, authors, journal, year, DOI. If you cannot find it, it does not go in.

— **Declare it** State which tool you used and what for, in a short note in your report. Disclosure is not a penalty; concealment is.

— **Cite it properly** APA 7 treats generative AI output as software: author (the company), year, name and version, plus the prompt in your method or an appendix.

— **The 20% rule** Turnitin similarity above 20% — whether from copying or from over-generated text — is returned for revision.

02

Finding Literature

Google Scholar and the İDU Library system

Finding Sources

01

Search like a researcher

Keywords and synonyms, not sentences. Boolean AND / OR / NOT, "exact phrase", and truncat* for word endings.

02

Filter deliberately

By date for currency, by author to follow a scholar, and by citation count to find what a field is built on.

03

Follow the network

'Cited by' moves forward in time; the reference list moves back. Connected Papers and Research Rabbit draw the map.

04

Export, don't retype

Pull APA 7 from Scholar into Zotero or Mendeley — then check it, because exported records are often wrong.

05

Use the library

İDU catalogue, e-journals, e-books and subscribed databases (ProQuest, EBSCOhost) reach what Scholar only shows you a paywall for.

06

Today's task

Search 'Artificial Intelligence in Education' or 'Sustainability'. Find one article and one book, export both in APA 7, and share them on Teams.

See you next week…

